A. Kamilka and the Sheep

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

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Kamilka has a flock of n sheep, the i-th of which has a beauty level of a_i. All a_i are distinct. Morning has come, which means they need to be fed. Kamilka can choose a non-negative integer d and give each sheep d bunches of grass. After that, the beauty level of each sheep increases by d.

In the evening, Kamilka must choose **exactly two** sheep and take them to the mountains. If the beauty levels of these two sheep are x and y (after they have been fed), then Kamilka's *pleasure* from the walk is equal to $\gcd(x, y)$, where $\gcd(x, y)$ denotes the greatest common divisor (GCD) of integers x and y.

The task is to find the maximum possible pleasure that Kamilka can get from the walk.

Input

Each test consists of several test cases. The first line contains one integer t ($1 \le t \le 500$), the number of test cases. The description of the test cases follows.

The first line of each test case contains one integer n ($2 \le n \le 100$), the number of sheep Kamilka has.

The second line of each test case contains n **distinct** integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the beauty levels of the sheep.

It is guaranteed that all a_i are distinct.

Output

For each test case, output a single integer: the maximum possible pleasure that Kamilka can get from the walk.

Example

Input
4 2 1 3 5 5 4 3 2 1 3 5 6 7 3 1 11 10

Output
2 4 2 10

Note

In the first test case, $d=1$ works. In this case, the pleasure is $\gcd(1+1, 1+3)=\gcd(2, 4)=2$. It can be shown that a greater answer cannot be obtained.

In the second test case, let's take $d=3$. In this case, the pleasure is $\gcd(1+3, 5+3)=\gcd(4, 8)=4$. Thus, for this test case, the answer is 4.